

Proposal to Investigate Luminary Green's Urban Design and its Impact on the UHI Effect and Social Utilization

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Background

The University of Central Florida (UCF) Downtown Campus, Creative Village, and Luminary Green were each designed by the City of Orlando to benefit the local community. Specifically, Luminary Green, a man-made “2.3-acre park... [with a] large, welcoming lawn,” was designed to facilitate everyday recreation and occasional outdoor events for the community (City of Orlando, 2021). Moreover, Luminary Green's current design may influence Downtown Orlando's year-round Urban Heat Island (UHI) effect.

The UHI effect is a phenomenon where developed urban areas experience “much warmer temperatures than nearby [less developed] rural areas” (National Aeronautics and Space Administration, 2025). In comparison to rural areas, urban areas have a plethora of heat-absorbing surfaces (cement, asphalt, dark roof, etc.) and scarce vegetation cover. For these reasons, urban areas often retain excess heat, creating considerable temperature differences between rural and urban areas. In the case of Orlando, temperature “differences between the... the city and... urban fringe [can differ] by as much as 14° F” (Winsberg & Simmons, n.d.). As reference, increased heat stress can directly induce “heat syncope, cardiovascular stress, thermal exhaustion, heat stroke and cardiorespiratory diseases” in urban populations (O'Malley et al., 2014). Furthermore, increased air temperatures can lead to the “formation of ground level ozone..., alteration of local climates... [,] change in local ecosystems... [, and] exacerbation of global warming,” impacting both urban populations and local environments (O'Malley et al., 2014). To reduce urban heat, planners must consider mechanisms to reduce the UHI effect.

To reduce the UHI effect, urban planners have begun focus on “vegetation restoration” within urban areas (Yu et al., 2025). Vegetation plays an important role in reducing the UHI effect through “evapotranspiration, shading, and albedo modification” (Yu et al., 2025). In short, vegetation generates water vapor and shade, cooling the Earth’s atmosphere. Furthermore, vegetation absorbs and reflects the sun’s light, preventing excess heat from being trapped in the Earth’s surfaces. However, the influence of vegetation on the UHI effect is not universal. The species (grass, trees, shrubs, etc.) and overall cover of vegetation can impact its ability to effectively reduce the UHI effect. For this reason, planners need to consider the effectiveness of urban vegetation designs, like Luminary Green, in reducing the UHI effect. Furthermore, if urban vegetation designs are ineffective in reducing the UHI effect, year-round social utilization of these spaces may be reduced.

Methodology

This project will entail a two-part analysis of (1) Luminary Green’s impact on Downtown Orlando’s year-round UHI effect and (2) the UHI effect’s impact on Luminary Green’s year-round social utilization. Once both parts are complete, all resulting data will be compiled into a final, interactive story map.

The first part of this analysis will showcase Luminary Green’s impact on Downtown Orlando’s year-round UHI effect. ArcGIS Pro (version 3.6) will be essential to this part of the analysis. Firstly, Landsat-8 “Band 4, Band 5, and Band 10” will be sourced (from the US Geological Survey) to generate data on Downtown Orlando’s year-round UHI effect (Jumari et al., 2023). Secondly, a Luminary Green and Downtown Orlando boundary will be sourced (from the US Census Bureau) to limit imagery to the area(s) of interest. Band 4 (Red) and 5 (Near Infrared) will be used to generate a “Proportion of Vegetation (Pv)” and “Land Surface

Emissivity (LSE)” layer (Jumari et al., 2023). Band 10 (Long Wavelength Infrared) will be used to generate a “Brightness Temperature (BT)” layer (Jumari et al., 2023). Together, the Pv, LSE, and BT layers will be combined to generate a “Land Surface Temperature (LST)” layer (Jumari et al., 2023). The LST layer will be used to generate a “Urban Heat Island (UHI)” layer (Jumari et al., 2023). The UHI layer will be used to calculate the difference in average urban heat across Luminary Green and Downtown Orlando. This process will be completed four times, utilizing imagery from the summer, fall, winter, and spring months, to calculate year-round differences.

The second part of this analysis will showcase the UHI effect’s impact on Luminary Green’s year-round social utilization. Physical travel to Luminary Green will be essential to this part of the analysis. Similar to the first part of this analysis, observations will occur in the summer, fall, winter, and spring months. The exact date of observation will be based on the imagery’s date of capture. One-hour observations will occur three times (morning, midday, and evening) each observation day. During these observations, quantitative counts of social utilization will be taken. Seasonal social utilization counts will be compared to the same season’s UHI effect determined in the first part of the analysis.

Unlike the prior part of this analysis, which relies on pre-existing, year-round US Census and USGS data, this part of the analysis aims to generate original, quantitative data of a year-round phenomenon. As reference, each season experiences varying levels of heat exposure due to variations in environmental factors (cloud cover, wind patterns, day length, etc.). These seasonal variations can reduce or induce the UHI effect. As such, seasonal variations in the UHI effect could result in varying levels of social utilization for each season. As such, a year-long study period will be required for this project.

Once both parts are complete, all data will be compiled into an interactive story map. ArcGIS StoryMaps will be essential to finalization of this project. Firstly, the interactive story map will provide four separate scenes that each showcase one of the four seasons (summer, fall, winter, and spring). In each scene, the UHI effect layer, which highlights both Downtown Orlando and Luminary Green, will be provided. Text descriptions, which compare the average temperatures of these two areas, will also be provided. Furthermore, text descriptions will highlight the average social utilization of Luminary Green for each scene.

Anticipated Outcomes

Overall Outcomes

This research will determine (1) Luminary Green's impact on Downtown Orlando's year-round UHI effect, and (2) the UHI effect's impact on Luminary Green's year-round social utilization.

Significance

- If results are positive, this project could suggest that Luminary Green's current design could be replicated in urban areas to reduce the UHI effect and increase social utilization.
- If results are negative, this project could suggest that Luminary Green's current design may need to be altered to efficiently reduce the UHI effect and increase social utilization.

Intended Audience

- The City of Orlando and UCF's urban planners.
- Urban planners from similar urban areas.

Project Benefits

- Updated knowledge on Downtown Orlando and Luminary Green's UHI effect.
- Insights on how specific urban vegetation impacts the UHI effect and social utilization.
- Greater insights into the year-round social utilization of UCF's open spaces.

References

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Timeline

Dates	Tasks
January 1, 2027 – January 8, 2027.	<u>Analysis #1: UHI Effect Luminary vs Downtown</u>

	Online data compilation (from USGS and US Census Bureau), extent limitation, and (4x) UHI layer generation.
January 1, 2027 – February 28, 2027	<u>Analysis #2: UHI Effect and Social Utilization</u> In-person ‘Winter’ observation and data recording: 1x day, 3x 1-hr long observations (morning, midday, evening).
March 1, 2027 – May 31, 2027	<u>Analysis #2: UHI Effect and Social Utilization</u> In-person ‘Spring’ observation and data recording: 1x day, 3x 1-hr long observations (morning, midday, evening).
June 1, 2027 – August 31, 2027	<u>Analysis #2: UHI Effect and Social Utilization</u> In-person ‘Summer’ observation and data recording: 1x day, 3x 1-hr long observations (morning, midday, evening).
September 1, 2027 – November 30, 2027	<u>Analysis #2: UHI Effect and Social Utilization</u> In-person ‘Fall’ observation and data recording: 1x day, 3x 1-hr long observations (morning, midday, evening).
December 1, 2027 – December 31, 2027.	<u>Final Deliverable: Interactive Story Map</u> Online compilation of data from Analysis #1 and Analysis #2 into a final interactive story map.

Budget

Item	Units and Cost	Total
Car Gas Reimbursement:	27.2 (full-trip miles) x 4 (trips) = 108.8 (miles) x \$0.445 (per mile) = \$4.84	\$4.84
Total		\$4.84